Unidirectional Security Gateway

COMP 4900K Group 4 Project

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Abstract

1 Background to Unidirectional Security Gateways

Protecting vital infrastructure from cyberattacks is crucial in today's networked digital environment. By establishing a physical barrier that permits data to move only in one direction—from secure operational technology (OT) networks to external networks—unidirectional security gateways, or USGs, have become a reliable method for stopping prospective cyberattacks from breaching sensitive systems. Usually, hardware elements like data diodes are used to enforce this unidirectional data flow, making sure that neither malware nor data may return to the network under protection.

In industrial control systems, where they enable the safe transfer of operational data to cloud services or enterprise networks without exposing essential systems to outside dangers, USGs are very important. USGs preserve the integrity and security of the OT environment while facilitating smooth data access for monitoring and analysis through server and device replication.

In addition to improving cybersecurity, USGs help businesses in a variety of sectors, including as manufacturing, transportation, and energy, comply with strict regulatory requirements. Adopting unidirectional gateways is a proactive strategy to safeguard critical services and infrastructure against potentially catastrophic cyber events as cyber threats continue to change.

2 Design & Implementation

A unidirectional gateway system based on the ubuntu operating system.

Unidirectional gateway consists of a unidirectional receive/transmit data module unit on a hardware integrated server, which contains a dual power supply module for continuous source power supply. It has an Ethernet switch port for data exchange and transmission between different devices, and I/O modules for input and output devices. In the SGX trusted computing environment within the data processing module and storage module for the use of data transmission security check mode for verification, storage of data and then forwarding/receiving data between the unit using a unidirectional optical fibre for connection.

The unidirectional data transmission platform can be realised in pass-through mode. In the pass-through mode, the data transmission pass-through application module in the system application level to build pass-through forwarding service, on the receiving data without any processing, the original in the receiving unit to transmit to the transmitting unit, the transmitting unit according to the routing for forwarding. Unidirectional data transmission platform hardware can be achieved to transmit data directly from the secure environment to the untrustworthy environment.

The unidirectional data transmission platform implements the security check mode. In the security check mode, the data transmission security check application module stores the data received in the receiving unit, filters the content according to the built-in algorithm, and performs error correction coding when the result is safe, and sends it to the sending unit unidirectionally after completing the coding. The data processing module of the sending unit carries out data error

correction and coding verification, and when the result is normal, the data is forwarded after local storage.

The structure of unidirectional data transmission platform is shown in Fig. 1, which realises two transmission modes: the first one is the transmission mode, which does not process the data and forwards the data directly in one direction. The second is the security inspection mode, which forwards the data after inspecting the data security, integrity and availability.

1). Receiving unit: it is used to receive the data sent by the equipment in the security environment and forward the data to the sending unit. 2). Sending unit: used to receive the forwarded data from the receiving unit, send the data from this unit outward. 3). Data exchange Ethernet port: used as an interface for data input/output of this unidirectional data transmission platform. 4). Power supply module. 5). I/O module. 6). Storage module: for local storage of received and forwarded data. 7) Data Processing Module: to filter the data. Checks for security, availability, integrity. 8) Unidirectional fibre optic module.

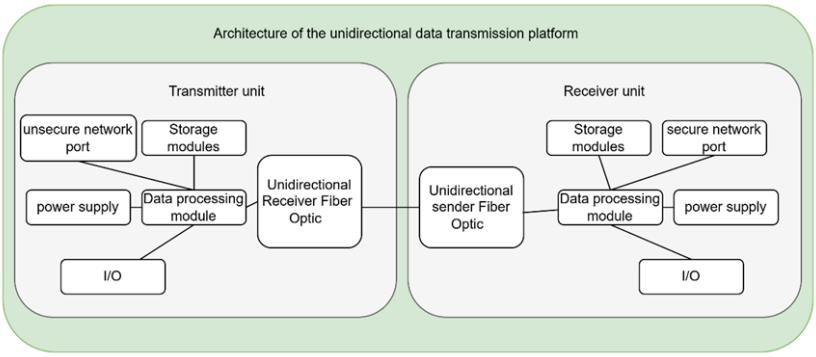

Fig. 1

Unidirectional transmission in pass-through mode

In the pass-through mode, the data transmission pass-through application module builds a pass-through forwarding application service at the system application level and does not perform any processing on the received data but transmits it unchanged from the receiving unit to the sending unit. The structure in this mode is shown in Fig. 2.

1). For cases that require pass-through forwarding, set the appropriate IP address and port at the receiving unit. 2). Set unicast or multicast in the receiving unit and the sending unit. 3). After the data exchange Ethernet port of the receiving unit receives the source data, forward the data to the unidirectional fibre module of this unit. 5). The unidirectional optical module of the receiving unit forwards the data to the unidirectional optical module of the sending unit. 6). The unidirectional fibre optic module sends the data to the insecure network port.

Unidirectional transmission in security mode is shown as Fig.3

1). The internal security server sends data to the receiving end. 2). The data processing module at the receiving end filters the data, uses error-correcting codes and signs the processed data. 3). Transmission of data: The processed data and the certification report are sent to the transmitting data processing module via a unidirectional fibre optic module. 4). The sending data processing module receives the data via the unidirectional fibre optic module. 5). The sending data processing module verifies the integrity of the data and the trusted identity of the sending unit. 6). The sending end data processing module decodes and corrects the data to recover the original data. 7). The transmitter data processing module sends the data to the external network.

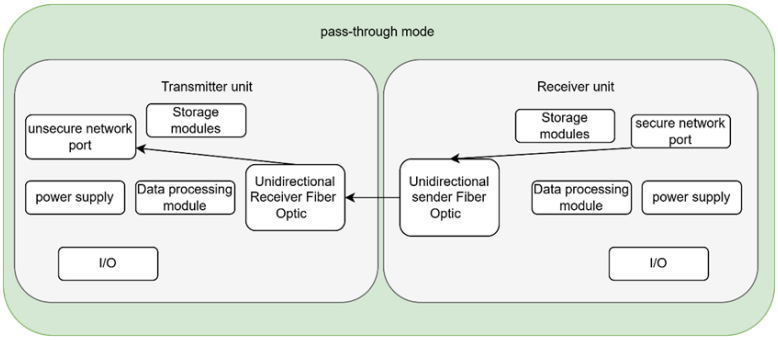

Fig.3

3 Security Analysis

A brief description of the role of a traditional one-way gateway

A traditional one-way gateway is a combination of software and hardware that securely replicates data from a high-security environment to an external network in a lower-security environment. The replicated applications can securely integrate with external applications (1). If this one-way gateway serves as the only information dissemination channel between the high-security and low-security environments and the one-way gateway works properly. The internal security network is protected from all online attacks from the external network (1). The one-way gateway sends data in such a way that the security system internally sends a replicated database to the outside through the one-way gateway when data is updated. Users on the external network will only interact with the replicated database (1). In this way it is possible to secure the internal network without affecting the normal interaction of external users.

Why use a traditional unidirectional gateway

Unidirectional gateways do not support bi-directional data transfer due to their physical layer. This results in many bi-directional links that are not legally required to be used when using a unidirectional gateway. This ultimately results in lower compliance costs and complexity for medium impact assets (1). This represents a reduction in the cost of deployment for unidirectional gateways. One-way gateways are also considered a replacement for firewalls (1). This represents that unidirectional gateways can replace firewalls and have good performance in certain contexts. When targeting nuclear power plants, unidirectional gateways also have less usage requirements compared to firewalls (1). This represents the advantage that unidirectional gateways are simpler and more secure to deploy than firewalls. Additionally, unidirectional gateways are often deployed to protect a portion of network connections that appear to require bidirectional communication (1).

Combination of SGX and traditional one-way gateway technology

Unidirectional gateways are generally used in critical industrial facilities and national infrastructures (2). This represents a strong importance of the data transmitted by unidirectional gateways. This means that not all data should be sent to external unsecured network environments. This means that we need a computational unit or chip to audit, filter and process the data using error correcting codes for the content entering and being sent by the unidirectional gateway. IN order to ensure the security of the computing unit, our team decided to introduce the SGX Trusted Computing Environment into the unidirectional gateway, which allows content auditing, filtering, and error-correcting coding algorithms to run securely in the Trusted Computing Environment. By allowing the algorithms to run in an Enclave, the security and integrity of the data processing is ensured. One-way gateways as a hardware device even in highly secured critical facilities still have the possibility of being recognised for physical attacks, SGX as a trusted computing environment can also resist physical attacks. This means that SGX can be combined with the physical isolation features of one-way gateways to further increase the security of one-way gateways. Unidirectional gateways limit the direction of data transmission, while SGX provides a trusted environment for data processing and algorithm execution. The combination of the two enables secure data processing without external influence.

What are the traditional network transmission security protocols that we design one-way gateways to solve?

Since the usage scenarios of unidirectional gateways are mainly in industries or infrastructures. However, industrial equipment does not necessarily support existing network transport security protocols (4). This means that it is not possible to securely transmit data by building network security protocols into industrial instruments and using soft-armour encryption. In addition, encrypting network traffic requires significant performance overhead (5). This represents the need to add a new piece of hardware by. Secure transmission of information to the outside world without using existing network transmission security protocols. To solve the problem of not being able to exist network transport security protocols and the need for performance. We chose to combine SGX technology with unidirectional network port technology. This combination of technologies has been proven to provide strong security and good performance without relying on complex cryptographic operations (4).

Threat Modelling

Trusted and untrusted components:

The trusted components of the device are Data Processing Module, Storage Modules, Unidirectional Fiber Optic, secure network port. the untrusted components of the device are, Unsecure Network Port, I/O. Data Processing Module and Storage Modules are considered trusted components because they run inside the enclave. Due to the physical unidirectional nature of Unidirectional Fiber Optic, data can only flow from the internal secure environment to the external. It is also considered a trusted component. Since the IO and Unsecure Network Port are ports that link to the external network. They are untrusted components.

Threat Model:

The first type of attacker attacks by actively sending data to the security server. This attacker may attack the internal server by injecting erroneous data, bypassing server rules, or affecting server availability. For this type of attacker, the physical unidirectional nature of a single gateway is a good solution to this attack. Since the outside cannot actively send data to the internal server. This attack naturally cannot pose a threat. The second type of attacker is to launch an attack on the

unidirectional gateway directly inside the system. Firstly, SGX safeguards the behaviour of privileged software reading the data stored in the Enclave. Thís represents if the attacker cannot read the data stored in the Enclave stored in the Enclave without authorisation through the IO on the one-way gateway device. The second way is that the attacker will intercept and modify the data output from within the data processing module. The local validation of the one-way gateway that we have designed can also confirm whether the data has been tampered with or not. The third way is to directly gain administrator privileges on the internal server and send sensitive data. This attack is countered by the data filtering algorithms running within SGX.

Attacks that cannot be countered:

The unidirectional gateway we have designed is not able to cope with side-channel attacks. This is because side-channel attacks affect the security of SGX and the algorithms running inside (4). Another type of attack that cannot be countered is when the attacker physically damages the gateway directly. This can lead to the disruption of the entire transmission link. This leads to service unavailability.

Why Choose to Incorporate the SGX Trusted Computing Environment on Both the Receiving and Sending Sides

In the sending unit section, SGX Enclave is used to validate incoming attestation reports, ensure that the data has not been tampered with in transit, and confirm the trusted identity of the sending unit. This process ensures that the data received by the external network is authenticated and complete. This prevents the data processing unit in the receiving section from being corrupted or the data from being tampered with or falsified during transmission. Ensure that a double authentication mechanism exists within the system. Using SGX on both the receiving and transmitting side enables a dual authentication mechanism. The receiving side ensures that the data is trusted before it enters the one-way gateway, and the sending side ensures that the data is trusted after it leaves the one-way gateway. This double assurance greatly improves the security and trustworthiness of the entire system and reduces the risks associated with a single point of failure. Secure against internal and external threats. The SGX on the receiving end protects the internal data processing process from insiders or malware; the SGX on the sending end prevents external attackers from falsifying or tampering with the data. The protection on both ends complements each other, providing comprehensive coverage of potential security threats. Reduced attack surface. Attackers need to break both the receiving and sending SGX to pose a threat to the entire transmission process, which is extremely difficult in practice, thus improving the overall security of the system.

Improved Solution: Fully Offline Authentication Process

As a unidirectional gateway in a highly secure environment, the data processing module incorporating the SGX Trusted Computing Environment is not able to send information to the outside world and perform trusted authentication. Our team chose to use a completely offline authentication process.

Firstly, a one-time verification of the SGX Enclave is performed by the device vendor or a third party in an online environment before the one-way gateway is shipped or before the device is initialised. After successful authentication, a trusted certificate is generated. The trusted certificate includes signature and device identity information. The trusted certificate is pre-stored in the storage of the receiving and sending units as a basis for subsequent verification. When internally secured data is received at the receiving end, the data is first processed using an algorithm. The algorithms are

stored in the trusted computing environment at the factory. When the computing module at the receiving end has verified all the data to be sent, it generates a local proof report. This report contains the measurements of the Enclave, the status labels of the data processing and is signed locally with a pre-configured device trusted certificate. This report is sent to the trusted computing module of the sending unit together with the data to be sent. The computing module of the sending unit verifies the proof report generated by the receiving unit by using a preconfigured trusted certificate. The validation includes whether the signature matches the content of the trusted certificate, whether the Enclave match and the data processing status contained in the report are trusted. If the validation is successful it continues to the external untrusted network. If the validation fails, the data is discarded and sending stops.

This offline validation mechanism allows for remote validation of SGX in cases where the computing module is not actively connected to the network.

4 Limitations

While the adoption of USGs and SGX technologies create a robust and secure approach for data transferring, it is still vulnerable to other forms of attack vectors. USGs focus on enforcing one-way data traffic while the implementation of SGX allows for secure processing and verification in a TEE. Our security goal is effective at blocking traditional network-based attacks and minimizing internal data intrusions, but it is not designed to protect against side channel attacks. For example, SGX is still susceptible to side channel attacks where attackers can infer sensitive data based on the Enclave's operations and behaviour. An attacker that has compromised the system where SGX present, where they can run processes outside the enclave and observe how it interacts with shared resources like CPU caches.

Furthermore, the implementation of SGX will increase the performance overhead due multiple factors. Creating and managing Enclaves, transitions to/from enclaves, excessive cache misses, and excessive enclave paging are the main factors that heavily create additional overhead (3). Because performance bottlenecks are common due to the trade-off of prioritizing security, using SGX with USGs will introduce more overhead and increase processing time for real-time system operations.

Our proposal's limitation in handling side-channel attacks, hardware vulnerabilities, and operational challenges underscore the need for additional mitigations. We can improve SGX to be more resilient to side channel exploits like integrating constant-time algorithms to eliminate timing leaks or minimize enclave interaction with shared hardware resources. Some side-channel exploits have been addressed through newer generations of Intel SGX so keeping your hardware up to date is also important. Implementing real-time detection mechanisms, and/or making sure you are following the best security practices will also help minimize the risk of these advanced exploits. To address these vulnerabilities, steps can be taken to strengthen the system against side-channel attacks by adapting hardware protections, practice secure coding design, and implement will ensure system that is robust and adaptable enough to emerging threats.

5 Conclusions

In an increasingly interconnected world, unidirectional security gateways, or USGs, have emerged as a crucial technological advancement for protecting sensitive systems. USGs successfully stop

cyber-attacks from penetrating vital infrastructure by facilitating one-way data flow via strong hardware components like data diodes and optical fiber modules. They guarantee safe and smooth data interchange while preserving the integrity of operational environments when implemented in industrial control systems and other crucial areas.

This project explored the design, implementation, and operational modes of USGs, specifically focusing on pass-through and secure verification modes. By combining innovative hardware designs with rigorous data security checks, these gateways provide a multifaceted defense mechanism against modern cyber threats. Their significance in sectors including manufacturing, transportation, and energy is further highlighted by their capacity to satisfy strict regulatory requirements.

Although USGs have many benefits, their drawbacks—such as their incompatibility with specific protocols and the possibility of bottlenecks in secure verification modes—highlight the necessity for continued study and improvement. To increase effectiveness and scalability, future research could concentrate on strengthening protocol support, streamlining data processing, and incorporating sophisticated threat detection techniques.

In conclusion, the adoption of USGs represents a proactive approach to cybersecurity, reinforcing the resilience of critical infrastructure against evolving cyber threats. By addressing existing challenges and continuing to innovate, USGs will remain a cornerstone of secure, reliable digital ecosystems.

Division of work

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Work is the same at the presentation

References

[1] Ginter, A. (2013). Unidirectional security gateways: Stronger than firewalls. *Proceedings of the ICALEPCS, San Francisco, CA, USA*, 6-11.

[2] Kim, B. K., Heo, Y. J., & Na, J. C. (2017, October). Design of unidirectional security gateway system for secure monitoring of OPC-UA data. In *2017 International Conference on Information and Communication Technology Convergence (ICTC)* (pp. 1287-1288). IEEE.

[3] (2016). Performance Considerations for Intel Software Guard Extensions (Intel SGX) Applications [White Paper]. Intel Corporation.

[4] Zhan, M., Li, Y., Yu, G., Zhang, Y., Li, B., & Wang, W. (2023). GUARDBOX: A high-performance middlebox providing confidentiality and integrity for packets. *IEEE Transactions on Information Forensics and Security*, *18*, 2413-2426.

[5] Trach, B., Krohmer, A., Gregor, F., Arnautov, S., Bhatotia, P., & Fetzer, C. (2018, March). Shieldbox: Secure middleboxes using shielded execution. In *Proceedings of the Symposium on SDN Research* (pp. 1-14)